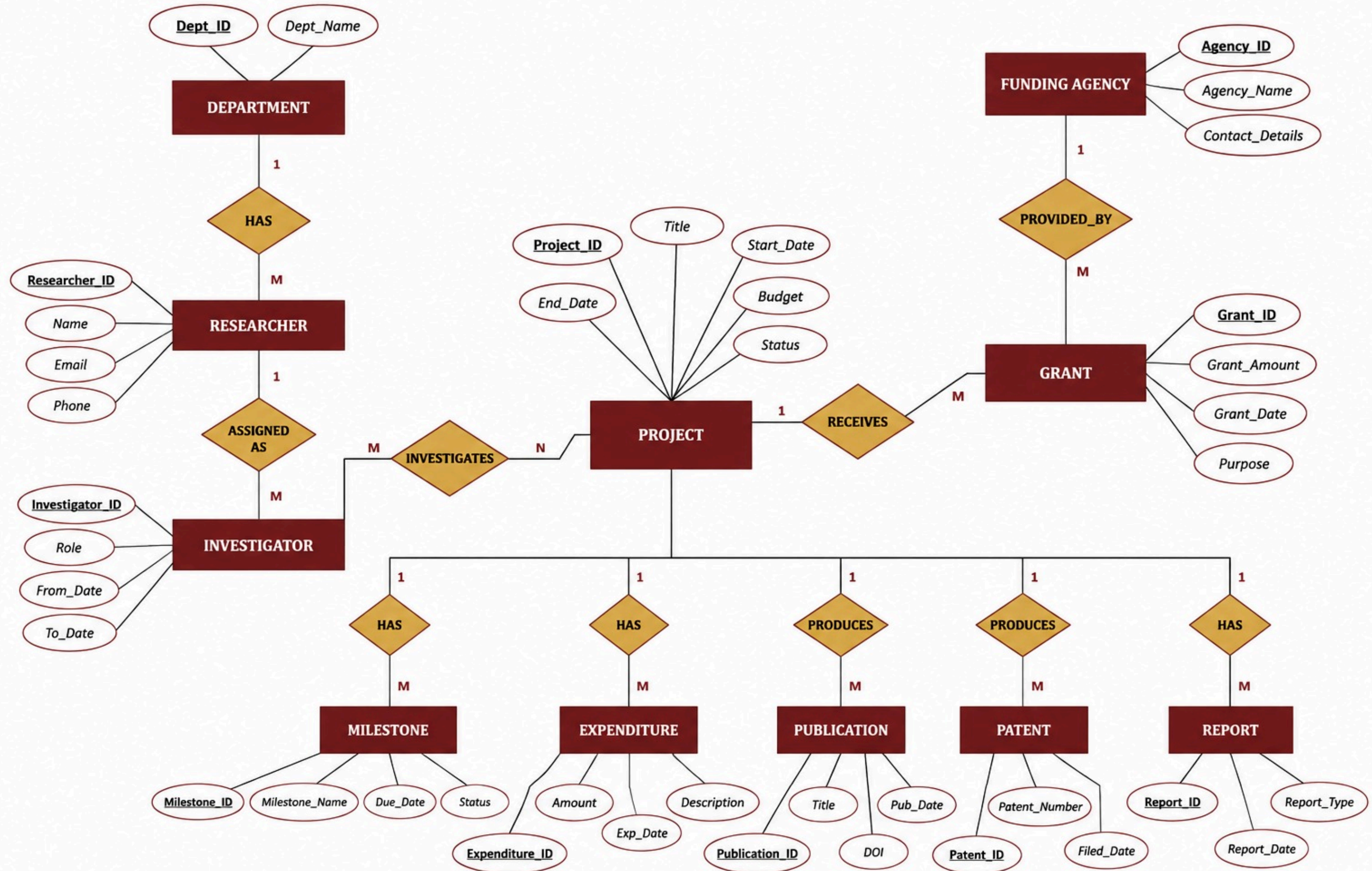


DBMS Assignment 1

Design and Implementation of a database management system for
Research Project, Publication and Grant Management System

Conceptual ER Diagram



Relational Schema

RELATION	ATTRIBUTES
DEPARTMENT	Dept_ID (PK), Dept_Name
RESEARCHER	Researcher_ID (PK), Dept_ID (FK), Name, Email, Phone
PROJECT	Project_ID (PK), Title, Start_Date, End_Date, Budget, Status
INVESTIGATOR	Investigator_ID (PK), Researcher_ID (FK), Project_ID (FK), Role, From_Date, To_Date
FUNDING AGENCY	Agency_ID (PK), Agency_Name, Contact_Details
RESEARCH GRANT	Grant_ID (PK), Project_ID (FK), Agency_ID (FK), Grant_Amount, Grant_Date, Purpose
MILESTONE	Milestone_ID (PK), Project_ID (FK), Milestone_Name, Due_Date, Status
EXPENDITURE	Expenditure_ID (PK), Project_ID (FK), Amount, Exp_Date, Description
PUBLICATION	Publication_ID (PK), Project_ID (FK), Title, Pub_Date, DOI
PATENT	Patent_ID (PK), Project_ID (FK), Patent_Number, Filed_Date
REPORT	Report_ID (PK), Project_ID (FK), Report_Type, Report_Date

Main Entities

Entity	Purpose
Department	Stores academic/research departments
Researcher	Stores researcher information
Project	Stores research project details
Investigator	Represents researcher involvement in projects
Funding Agency	Stores organizations providing funding
Grant	Stores project funding details
Milestone	Tracks project milestones
Expenditure	Records project expenses
Publication	Stores research publications
Patent	Stores project-related patents
Report	Stores project/research reports



Thank You